

NITHIN G KRISHNAN

Aeronautical Engineer

Trivandrum, Kerala

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Profile

Recent Aeronautical Engineering graduate with academic and practical exposure in Computational Fluid Dynamics (CFD), external aerodynamics, CAD-based design, and aerodynamic performance analysis. Experienced in CFD simulations using ANSYS Fluent, aerofoil and wing analysis, and engineering design using CATIA and SolidWorks. Skilled in interpreting aerodynamic flow behaviour, pressure distribution, and performance parameters through simulation studies. Seeking an entry-level role in CFD, Aerodynamics, or Design Engineering.

Education

APJ Abdul Kalam Technological University

2022 – 2026

Bachelor of Technology (B.Tech) – Aeronautical Engineering

Cumulative Grade Point Average (CGPA): 7.46

Technical Skills

Computational Fluid Dynamics

- External aerodynamics and wind flow simulation
- Boundary layer analysis and external flow modelling
- Lift, drag and moment coefficient evaluation
- Pressure, velocity and turbulence analysis

Simulation, CAD and Analysis Tools

- ANSYS Fluent – CFD simulations and aerodynamic analysis
- CATIA – 3D CAD modelling and component design
- SolidWorks – Mechanical and aerodynamic design
- MATLAB – Numerical computation and engineering analysis
- ANSYS – Geometry preparation and simulation support
- Microsoft Excel – Data analysis and result interpretation

Programming

- C Programming Fundamentals
- MATLAB Programming Fundamentals

Experience

Industry Internship – Air India Engineering Services Limited (AIESL)

- Applied academic engineering knowledge to practical aircraft maintenance operations.
- Worked with Tool Store, Wheels and Brakes, and Material Planning departments.
- Gained exposure to aircraft maintenance procedures, tooling systems, and engineering coordination.
- Developed understanding of MSDS documentation and aircraft safety compliance practices.

Academic Projects

Effect of Riblets on the Performance of an Aircraft Wing

- Conducted CFD simulations to evaluate drag reduction using riblet surface technology.
- Analyzed aerodynamic characteristics using pressure and velocity contour plots.
- Compared lift and drag performance to identify aerodynamic improvements.

Hybrid (Reflex + Cambered) Aerofoil Design for Tailless UAVs

- Designed and analyzed a hybrid aerofoil configuration for tailless UAV applications.
- Evaluated lift-to-drag ratio, pressure distribution, and stability characteristics.
- Studied flow separation, boundary layer behaviour, and aerodynamic coefficients using CFD.

CFD Reporting Experience

Prepared structured CFD simulation reports including geometry preparation, meshing strategy, boundary condition setup, solver configuration, validation methodology, result interpretation, and engineering conclusions. Developed technical documentation supported by aerodynamic contour plots, graphs, and comparative performance analysis.

Certifications

Professional Certifications

- ANSYS Certification – Advanced simulation and analysis
- NPTEL Certification – Introduction to Experiments in Flight
- Programming in C – Infosys Springboard

Industry Experience

- Internship Certificate – Air India Engineering Services Limited (AIESL)

Specialized Training

- UAVs and Drones Workshop – Design and operational principles
- Industrial Automation Workshop – Manufacturing processes
- Universal Human Values – AICTE

Soft Skills

- Analytical and structured problem solving
- Technical report writing and documentation
- Team collaboration in engineering environments
- Attention to detail in simulation and data analysis
- Fluent in English communication